

What Is Machine Learning (ML)? Definition and Examples

March 17, 2026

How can computers learn without being explicitly programmed? This is the core question that drives the growing field of machine learning (ML), a type of artificial intelligence that is increasingly integrated into our daily lives.

While ‘machine learning’ is a buzzword, many people don’t fully grasp what exactly it means. How do machine learning algorithms work? What role does machine learning play in AI? Even systems designed to think in purely mathematical terms can take on human-like traits, including bias, making [data ethics and transparency](#) essential throughout the model development process.

We will explore the fundamental question, “What is ML?” by providing insight into how machine learning works, its development, and what the future holds for this evolving technology.

Key Takeaways

- Machine learning is a subset of AI concerned with training models to allow computers to mimic human thought and decision making without explicit programming.
- The most common types of ML are supervised learning (learning via labeled data), unsupervised learning (learning via unlabeled data), and reinforcement learning (learning via a reward and punishment response).
- While ML drives powerful benefits like automation, advanced pattern recognition, and personalization, it also has risks, including algorithmic bias and data privacy concerns.

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Definition of Machine Learning

Machine learning (ML) is concerned with training models to allow computers to mimic human thought and decision making without explicit programming.

Using statistical methods, computers analyze datasets and identify patterns within them to predict outcomes or produce outputs ([view a visual of machine learning via R2D3](#)). Through model training using techniques like gradient descent and hyperparameter tuning, we optimize these predictions.

Machine Learning (ML) vs. Artificial Intelligence (AI)

Machine learning and artificial intelligence (AI) are closely related: ML is a subset of artificial intelligence and is the driving technical force behind most modern AI systems.

AI is the umbrella term including any technique dedicated to trying to perform tasks typically associated with human intelligence.

Machine learning can be considered a sub-field of AI and is dedicated to developing systems that extract patterns from data.

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How Does Machine Learning Work?

Traditional programming can be thought of like baking — a recipe, or “program,” details the specific ingredients and step-by-step instructions used to achieve the desired output. Now, imagine writing step-by-step instructions for complex tasks such as “recognize the written letter” or “predict what TV shows this user would enjoy.” These are difficult or time-consuming to describe to a computer.

With machine learning, computers teach themselves the instructions. The computer studies training data, identifies patterns and relevant parameters, and uses those inferences to test its predictions on new data it hasn't studied before.

With each iteration of this process, the error rate is reduced until the error rate is sufficiently low.

Three Components of a Machine Learning Model

There are a variety of different ML model architectures and types. However, the typical supervised machine learning algorithm consists of roughly three components:

1. A Decision Process

A decision process is a recipe of calculations or other steps that takes in the data and “guesses” what kind of pattern your algorithm is looking to find.

2. An Error Function

An error function is a method of measuring how good the guess was by comparing it to known examples (when they are available). Did the decision process get it right? If not, how do you quantify “how bad” the miss was?

3. An Updating or Optimization Process

An updating or optimization process is a method in which the algorithm looks at the miss and then updates how the decision process comes to the final decision, so next time the miss won’t be as great.

Example of How ML Models Work

For example, if you’re building a movie recommendation system, you can provide information about users and their watch history as the training data. The algorithm will take that input and learn how to return an accurate output: movies that users will “enjoy” (e.g., rate highly or watch all the way through).

Some inputs could be:

- Movies users watched and rated highly
- Percentage of movies users watch in each genre
- Whether a user favors a certain actor or director
- Time of day that certain movies are watched

The algorithm’s job is to adjust the weights to map the input to the output and minimize the errors. If the algorithm gets it right, the weights it used stay the same. If it gets a movie wrong, the weights that led to the wrong decision get adjusted so it’s less likely to make that kind of mistake again.

The machine learning algorithm processes the training set, the set of training examples, autonomously, aiming to reduce the error rate as it learns from the data it analyzes. This iterative nature of learning is both unique and valuable because it occurs without human intervention — empowering the algorithm to uncover hidden insights without being specifically programmed to do so.